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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	TEAM UP
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 12 - Responsible Consumption and Production
Project / Solution Name	B-WARN
Name (Leader)	JOVAN JEAN LIM YUE HONG
Email & Phone (Leader)	017-6337886
University Affiliation(s)	UNIVERSITI TEKNOLOGI MALAYSIA
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized enterprises (SMEs), especially convenience stores and small food retailers often face difficulties managing inventory, monitoring expiry dates and maintaining accurate financial records. Many rely on manual stock checks, spreadsheets or disconnected systems hence increasing the risk of human error and inefficient operations.

One major challenge is identifying products and connecting them to their expiry dates. In this context, products that pass their expiry date will pose a loss for the business and also create unnecessary waste .Therefore, our team believed that selling near expiry products at a lower profit margin is more optional to the business and also reduces the total amount of waste.

SME owners especially those without prior business experience might miss timed promotions such as doing price discounts too late or too early. Besides that, manually changing all the products in the company database is also an extremely time consuming and extensive activity.

Along with that, the environmental issues involved is that expired food or products cannot be sold to customers and the only way is to throw them out. Another social issue that arises is the misuse of human capital. Workers will be better off doing productive tasks that generate revenue for the business that performing repetitive price changing

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Business Survival is a web-based SME management platform designed to automate inventory monitoring, support promotional decisions and simplify financial tracking.

The system allows business owners to register products, receive stock, monitor individual inventory batches, set selling prices and identify products approaching expiry. When products are eligible for

promotion, the owner can select the relevant batches and generate a promotional recommendation through our integrated ai promotion generator

Business Survival uses a hybrid AI approach. Python-based business rules first validate inventory eligibility and recommend discounts according to the remaining shelf life. The backend then integrates Google's Gemini generative AI model to produce promotional titles, email subjects and marketing messages using the selected products' names, quantities, expiry dates and discount information.

The owner can review and edit the generated content before saving a draft. Approval and promotion simulation are separate actions, ensuring that important decisions remain under human control. Simulation demonstrates the promotional workflow without sending actual customer emails.

Additional features include financial analytics, sales and expense recording, automatic inventory deduction following recorded sales and an activity history for accountability.

The application is built using React and TypeScript on Replit, with a Python FastAPI backend hosted on Render and Supabase for authentication and database management. Google Gemini is integrated through the backend API which keeps the AI API key outside the browser.

By combining connected business information, rules-based automation and generative AI, Business Survival aims to make everyday business management more efficient and accessible for SMEs.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend

- React and TypeScript
- Replit for development and frontend deployment
- Responsive web interface for desktop and mobile devices

Backend

- Python
- FastAPI
- Render cloud hosting
- REST API endpoints connecting the frontend to business logic

Database and Authentication

- Supabase
- PostgreSQL database
- User authentication
- Server-side database operations and transaction validation

Artificial Intelligence

- Google Gemini API
- Google Gen AI Python SDK
- Hybrid AI generation with rule-based inventory and discount validation

Version Control

- GitHub for backend source code and change tracking

System Architecture

User / Business Owner

↓

React + TypeScript Frontend (Replit)

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Authenticated REST API Requests

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Python FastAPI Backend (Render)

↙ ↘

Supabase Database Google Gemini API

Inventory, sales, Promotional text

prices, promotions generation

and activity records

↓

Validated Response to Frontend

↓

Owner Reviews, Saves and Approves

↓

Simulated Outcome and Activity History

How the system works

Inventory records, expiry dates and quantities are retrieved from Supabase through the authenticated backend. Python business rules identify eligible promotion batches and calculate suggested discounts. Google Gemini uses validated product facts to generate promotional content.

The system also automatically calculates total sales revenue, recorded expenses, net profit and profit margin from saved financial transactions. Business owners can view these figures through financial

dashboards with daily, monthly and yearly filters. When a product sale is recorded, the backend automatically deducts the sold quantity from eligible inventory batches, improving stock accuracy.

Business owners review AI-generated promotional recommendations before making the final decision. Approved promotions can be simulated, and workflow actions are recorded in Activity History for accountability.

The financial calculations use recorded transaction amounts. Product costs must be entered as expenses to be reflected in the reported net profit.

Official SDC Sponsor-Provided GenAI API or Cloud Platform

No official sponsor-provided GenAI API or cloud platform was used. The project uses Replit, Render, Supabase and Google Gemini independently.

Project Links

- Website: https://business-survival.replit.app
- Backend GitHub: https://github.com/09028jovanlim-pixel/business-survival-backend

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Business Survival follows a user-centred development approach focused on the daily operational difficulties experienced by small business owners.

Empathise and Define: The project focuses on common SME challenges, particularly manual inventory tracking, preventable product expiry, time-consuming promotion preparation and disconnected financial records.

Ideate: The proposed solution combines inventory management, expiry alerts, AI-assisted promotional content and financial monitoring into one platform rather than requiring business owners to use multiple separate tools.

Prototype: A functional web application was developed using React, TypeScript, Python FastAPI, Supabase, Google Gemini and Render. The prototype includes real database operations, authenticated access, inventory validation and human-controlled promotional workflows.

Test and Improve: Development testing covered duplicate product prevention, inventory updates, expiry eligibility, stock deduction, price persistence, transaction recording and promotion approval behaviour. Interface improvements were made to improve responsiveness and usability.

Deployment Strategy: The application uses cloud-hosted services to support accessibility and simpler deployment. Future work will focus on multi-company data isolation, stronger role-based permissions, improved AI reliability, automated promotional delivery and further usability testing with SME owners.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary SDG: SDG 12 — Responsible Consumption and Production** **Target 12.3:** By 2030, halve per-capita global food waste at retail and consumer levels and reduce food losses along production and supply chains. Business Survival supports this target by helping food retailers identify products approaching expiry and take timely action through inventory-based promotional recommendations. **Secondary alignment: SDG 8 — Decent Work and Economic Growth** The project also supports productivity and SME innovation by reducing repetitive manual inventory and administrative tasks.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?*	**Prototype demonstration baseline:** One monitored Milk inventory batch contained **99 units** approaching expiry on 29 October 2026.

Metric Category	**Your Response**
e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."	These 99 units represent identified **at-risk inventory**, not confirmed food waste. Without a timely response, some stock could remain unsold and eventually expire. For Phase 2 evaluation, baseline performance will be established by tracking units that expire unsold during a defined observation period. **Baseline measurement:** ● At-risk stock observed in the prototype: 99 units in one eligible batch. ● Actual historical spoilage rate: Not yet measured. ● SME field-validation baseline: To be established during pilot testing.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary KPI: Percentage reduction in units of inventory expiring unsold.** Measurement: Percentage reduction = ((Baseline expired-unsold units − Post-implementation expired-unsold units) ÷ Baseline expired-unsold units) × 100% Supporting indicators: ● Number of inventory units identified before expiry. ● Number of eligible batches reviewed for promotion. ● Number of promotional recommendations approved. ● Time required to prepare a promotional message. ● Value of stock potentially recovered through earlier action. These KPIs will distinguish actual measured results from projected benefits.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Target: Achieve a 20% reduction in inventory units expiring unsold during a controlled SME pilot**, compared with a baseline observation period. Business Survival aims to achieve this by detecting approaching expiry dates, generating timely promotional recommendations and reducing the time owners spend preparing promotional content. The target is a proposed performance goal, not an already proven outcome. Phase 2 testing would measure actual stock movement, unsold expired quantities and the effectiveness of the promotional workflow before reporting verified environmental benefits. **Reference:** United Nations Sustainable Development Goals — Goal 12, Target 12.3 https://sdgs.un.org/goals/goal12

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

[] Yes [/] No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[/] Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Gemini-3.5-flash-lite	**Pre-trained AI model**	**Proprietary (Google Gemini API Terms)**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

[]Yes [/] No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

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24/10/2026